

Kshitij Agarwal

Data Scientist

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Professional Summary

Data scientist with over 10 years of experience leveraging an engineering foundation to solve complex challenges through probabilistic and statistical modeling. Proven track record across domains:

- Built a mathematical model of human neurological posture control.
- Designed real-time closed-loop algorithms running **24/7** on industrial drilling rigs to detect dysfunctions, saving millions of dollars.
- Architected enterprise data quality automation scanning over **1,000,000** banking datasets.
- Partnered with leading AI labs (OpenAI, Microsoft via Turing) delivering **RLHF** to improve **LLM Python code** generation and agentic tool-calling workflows.

Technical Skills

- **Languages:** Python, SQL, R, MATLAB, Java
- **Data & ML / AI:** Machine Learning, Statistical Modeling, RLHF, Agentic AI, PySpark, Pandas, DB2, BigQuery, Hadoop
- **Tools & Platforms:** Google Cloud Platform (GCP), Airflow, Docker, Git, CLI, iOS Development
- **Spoken Languages:** English (Proficient), Hindi (Native)

Relevant Experience

Senior Developer

Turing Enterprises (*Remote*)

Jun 2025 – Dec 2025

- **LLM Code Generation Enhancement (OpenAI / Microsoft):**
 - Addressed code hallucination and syntax errors in early frontier models (GPT-4 and o1).
 - Designed complex coding scenarios and evaluated outputs with rigorous RLHF feedback, directly improving the generation of robust, production-grade Python code.
- **Agentic AI Reasoning & Tool Calling (OpenAI / Microsoft):**
 - Targeted multi-step reasoning bottlenecks and API/tool-calling failures on complex tasks (e.g., booking workflows, long-context reviews).
 - Supervised and rectified model outputs via RLHF protocols, empowering subsequent model revisions to handle autonomous multi-step agent actions.

Manager, Data Quality and Innovations

Bank of Nova Scotia (*Toronto, ON*)

Jun 2022 – Jul 2024

- **Data Quality Automation & Pipeline Architecture:**
 - Modernized legacy manual validation routines across more than 1,000,000 internal banking datasets.
 - Architected PySpark ETL pipelines aggregating data from Hadoop, DB2, GCP, and local storage.
 - Deployed automated data profiling alongside a Pandas-based anomaly detection framework incorporating SME feedback loops for remediation.
 - Cut manual processing costs by **\$26 million**, increased data management operational productivity by **40%**, and ensured data integrity for downstream AI products.

Research Engineer

Pason Systems Inc. (*Calgary, AB*)

Aug 2018 – Jun 2022

- **Real-Time Drilling Dysfunction Detection & Mitigation Algorithm:**

- Designed an autonomous, closed-loop real-time time-series algorithm in Java to mitigate equipment damage while maximizing drilling speed.
- Achieved a **99% dysfunction detection rate** and drove a **10% increase** in commercial product adoption.
- **Rotary Speed Optimization for Rate-of-Penetration (ROP):**
 - Collaborated with ExxonMobil to develop a Java-based algorithm optimizing drill bit rotary speed (RPM) in tandem with weight-on-bit (WOB).
 - Validated the system during active field trials across commercial rigs.
- **Predictive ROP Limiting Algorithm:**
 - Built a proactive predictive model in Java that dynamically capped ROP prior to dysfunction onset.
- **In-House MATLAB Simulation Platform:**
 - Created a closed-loop drilling simulator in MATLAB, reducing dependency on physical QA hardware racks and significantly shortening R&D release cycles.

Research Assistant

University of Alberta (*Edmonton, AB*)

Sep 2016 – Sep 2018

- **Theoretical Model for Human Postural Control:**
 - Investigated disrupted neuromuscular pathways caused by stroke and neurological disorders during seated balance.
 - Executed clinical biomechanical trials, conducted digital signal processing, and formulated a theoretical mathematical model explaining neural control of human upright sitting posture.

Other Experience

- **Senior Consultant** | EXL Services (*Gurgaon, HR*) Feb 2025 - Jun 2025
Implemented automated data quality pipelines with proactive anomaly detection to minimize manual oversight.
- **Manager** | Himalaya Developers (*Rishikesh, UK*) Jun 2024 – Feb 2025, Dec 2025 - Aug 2026
Directed end-to-end execution of two residential projects valued at \$3.2M over 18 months (procurement, licensing, civil engineering teams, financial modeling, and sales).
- **iOS Developer** | Pakshi Inc. (*Calgary, AB*) Jan 2020 – Jan 2022
Engineered and shipped two native iOS applications (*More ToLife* and *RoboTask*), managing launch pipelines and digital acquisition strategies.

Education

- **Master of Data Science (MDS)**
University of British Columbia, Vancouver, BC 2026 – Ongoing
- **Master of Science (M.Sc.)**
University of Alberta, Edmonton, AB 2016 – 2018
- **Bachelor of Technology (B.Tech. with Double Major)**
Indian Institute of Technology (IIT) Kanpur, Kanpur, UP 2011 – 2016

Patents & Credentials

Patent

- [US11499411B2](#)

Certifications

- [Advanced Agentic AI](#)
- [LangChain for LLM Development](#)
- [Machine Learning \(Stanford / Coursera\)](#)
- [Machine Learning Engineer Certificate](#)
- [Google Cloud Big Data and Machine Learning Fundamentals](#)
- [Exploring and Preparing your Data with BigQuery](#)